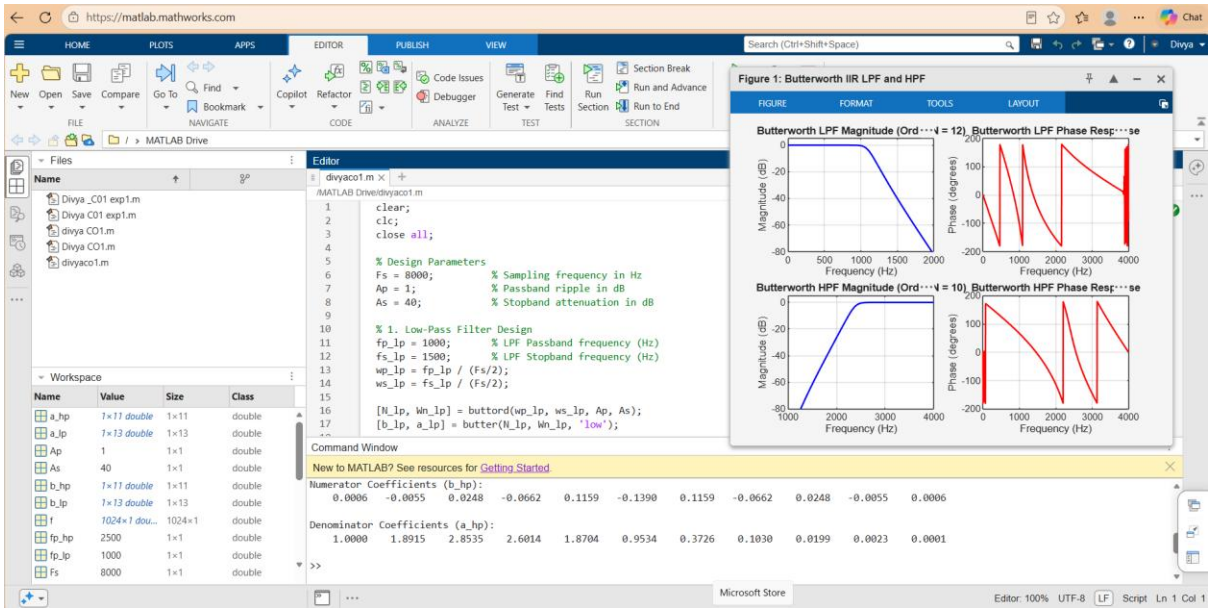

https://matlab.mathworks.com
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FILE
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CODE
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TEST
SECTION
MATLAB Drive
Files
Name
Divya _C01 exp1.m
Divya C01 exp1.m
divya CO1.m
Divya CO1.m
divyaco1.m
Workspace
Name
Value
Size
Class
a_hp 1×11 double 1×11 double
a_lp 1×13 double 1×13 double
Ap 1 1×1 double
As 40 1×1 double
b_hp 1×11 double 1×11 double
b_lp 1×13 double 1×13 double
f 1024×1 dou... 1024×1 double
fp_hp 2500 1×1 double
fp_lp 1000 1×1 double
Fs 8000 1×1 double
Editor
divyaco1.m
/MATLAB Drive/divyaco1.m
1 clear;
2 clc;
3 close all;
4
5 % Design Parameters
6 Fs = 8000; % Sampling frequency in Hz
7 Ap = 1; % Passband ripple in dB
8 As = 40; % Stopband attenuation in dB
9
10 % 1. Low-Pass Filter Design
11 fp_lp = 1000; % LPF Passband frequency (Hz)
12 fs_lp = 1500; % LPF Stopband frequency (Hz)
13 wp_lp = fp_lp / (Fs/2);
14 ws_lp = fs_lp / (Fs/2);
15
16 [N_lp, Wn_lp] = buttord(wp_lp, ws_lp, Ap, As);
17 [b_lp, a_lp] = butter(N_lp, Wn_lp, 'low');
Command Window
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Numerator Coefficients (b_hp):
0.0006 -0.0055 0.0248 -0.0662 0.1159 -0.1390 0.1159 -0.0662 0.0248 -0.0055 0.0006
Denominator Coefficients (a_hp):
1.0000 1.8915 2.8535 2.6014 1.8704 0.9534 0.3726 0.1030 0.0199 0.0023 0.0001
>>
Figure 1: Butterworth IIR LPF and HPF
FIGURE
FORMAT
TOOLS
LAYOUT
Butterworth LPF Magnitude (Ord···N = 12)
Butterworth LPF Phase Resp···se
Magnitude (dB)
Phase (degrees)
Frequency (Hz)
Butterworth HPF Magnitude (Ord···N = 10)
Butterworth HPF Phase Resp···se
Microsoft Store
Editor: 100% UTF-8 LF Script Ln 1 Col 1